

Администрирование локальных сетей

Лабораторная работа №6

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Цели и задачи работы

Настроить межвлановую маршрутизацию с использованием технологии Router-on-a-Stick и изучить процесс передачи пакетов в сети.

Ход выполнения работы

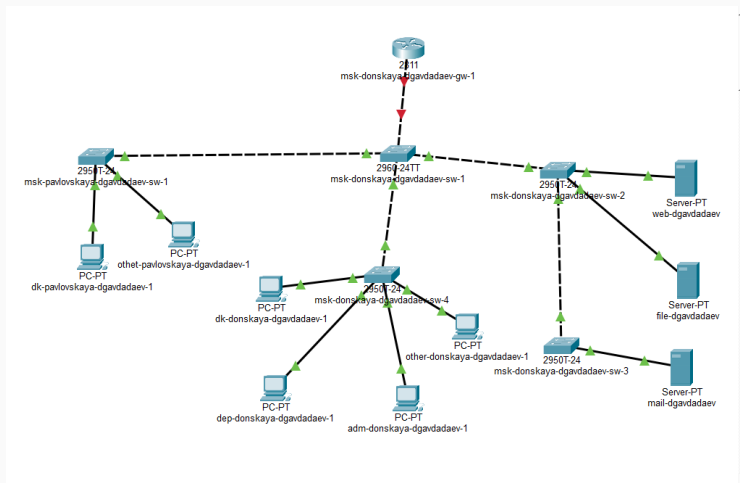


Рис. 1: Топология сети

Первоначальная настройка маршрутизатора

```
Router>enab
Router#conf term
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#host msk-donskaya-dgavdadaev-gw-1
msk-donskaya-dgavdadaev-gw-1(config)#line vty 0 4
msk-donskaya-dgavdadaev-gw-1(config-line)#pass cisco
msk-donskaya-dgavdadaev-gw-1(config-line)#login
msk-donskaya-dgavdadaev-gw-1(config-line)#line cons 0
msk-donskaya-dgavdadaev-gw-1(config-line)#pass cisco
msk-donskaya-dgavdadaev-gw-1(config-line)#login
msk-donskaya-dgavdadaev-gw-1(config-line)#enab sec cisco
msk-donskaya-dgavdadaev-gw-1(config)#serv pass
msk-donskaya-dgavdadaev-gw-1(config)#user admin priv 1 sec cisco
msk-donskaya-dgavdadaev-gw-1(config)#ip domain-name donsкаya.rudn.edu
msk-donskaya-dgavdadaev-gw-1(config)#crypto key gen rsa
The name for the keys will be: msk-donskaya-dgavdadaev-gw-1.donskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-dgavdadaev-gw-1(config)#line vty 0 4
*Mar 1 0:6:22.794: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:6:22.794: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-dgavdadaev-gw-1(config-line)#trans in ssh
msk-donskaya-dgavdadaev-gw-1(config-line)#
```

Рис. 2: Настройка маршрутизатора

```
msk-donskaya-dgavdadaev-gw-1(config)#
msk-donskaya-dgavdadaev-gw-1(config)#int f0/0
msk-donskaya-dgavdadaev-gw-1(config-if)#no shut

msk-donskaya-dgavdadaev-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

msk-donskaya-dgavdadaev-gw-1(config-if)#
msk-donskaya-dgavdadaev-gw-1(config-if)#int f0/0.2
msk-donskaya-dgavdadaev-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.2, changed state to up

msk-donskaya-dgavdadaev-gw-1(config-subif)#enc dot1q 2
msk-donskaya-dgavdadaev-gw-1(config-subif)#ip addr 10.128.1.1 255.255.255.0
msk-donskaya-dgavdadaev-gw-1(config-subif)#descr management
msk-donskaya-dgavdadaev-gw-1(config-subif)#
msk-donskaya-dgavdadaev-gw-1(config-subif)#int f0/0.3
msk-donskaya-dgavdadaev-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.3, changed state to up

msk-donskaya-dgavdadaev-gw-1(config-subif)#enc dot1q 3
msk-donskaya-dgavdadaev-gw-1(config-subif)#ip addr 10.128.0.1 255.255.255.0
msk-donskaya-dgavdadaev-gw-1(config-subif)#desc servers
msk-donskaya-dgavdadaev-gw-1(config-subif)#
```

Рис. 3: Настройка VLAN-интерфейсов

Конфигурация всех VLAN

```
msk-donskaya-dgavdadaev-gw-1
Physical Config CLI Attributes
IOS Command Line Interface
!
!
!
!
!
interface FastEthernet0/0
no ip address
duplex auto
speed auto
!
interface FastEthernet0/0.2
description management
encapsulation dot1Q 2
ip address 10.128.1.1 255.255.255.0
!
interface FastEthernet0/0.3
description servers
encapsulation dot1Q 3
ip address 10.128.0.1 255.255.255.0
!
interface FastEthernet0/0.101
description dk
encapsulation dot1Q 101
ip address 10.128.3.1 255.255.255.0
!
interface FastEthernet0/0.102
description departments
encapsulation dot1Q 102
ip address 10.128.4.1 255.255.255.0
!
interface FastEthernet0/0.103
description adm
encapsulation dot1Q 103
ip address 10.128.5.1 255.255.255.0
!
interface FastEthernet0/0.104
description other
encapsulation dot1Q 104
ip address 10.128.6.1 255.255.255.0
!
interface FastEthernet0/1
no ip address
duplex auto
speed auto
shutdown
!
```


Проверка связности

```
dk-pavlovskaya-dgavdadaev-1
Physical Config Desktop Programming Attributes
Command Prompt
Reply from 10.128.3.5: bytes=32 time<1ms TTL=128
Reply from 10.128.3.5: bytes=32 time<1ms TTL=128

Ping statistics for 10.128.3.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.128.4.5

Pinging 10.128.4.5 with 32 bytes of data:

Request timed out.
Reply from 10.128.4.5: bytes=32 time<1ms TTL=127
Reply from 10.128.4.5: bytes=32 time=3ms TTL=127
Reply from 10.128.4.5: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.4.5:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 1ms

C:\>ping 10.128.5.5

Pinging 10.128.5.5 with 32 bytes of data:

Request timed out.
Reply from 10.128.5.5: bytes=32 time<1ms TTL=127
Reply from 10.128.5.5: bytes=32 time<1ms TTL=127
Reply from 10.128.5.5: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.5.5:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.128.6.5

Pinging 10.128.6.5 with 32 bytes of data:

Request timed out.
Reply from 10.128.6.5: bytes=32 time<1ms TTL=127
Reply from 10.128.6.5: bytes=32 time=2ms TTL=127
Reply from 10.128.6.5: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.6.5:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>|
```

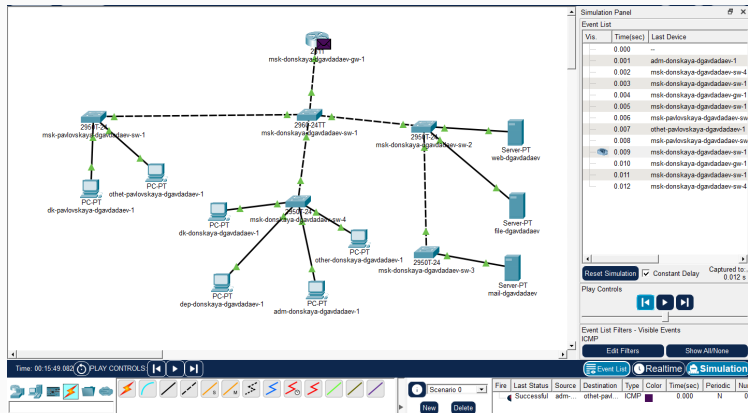


Рис. 6: Simulation

Анализ структуры пакета

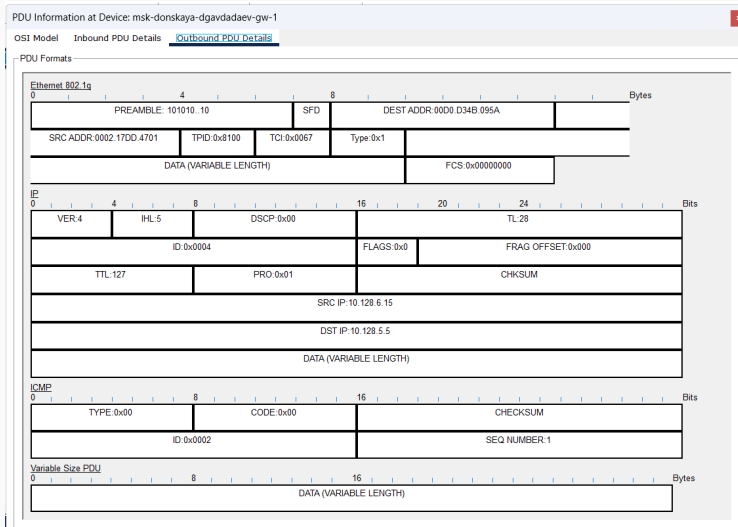


Рис. 7: Структура пакета

Выводы

В ходе выполнения лабораторной работы:

- реализована межвлановая маршрутизация
- настроена технология Router-on-a-Stick
- обеспечена связность между VLAN
- изучен процесс передачи пакетов
- проанализированы заголовки сетевых протоколов